

# Introduction to SQL Developer

COM 736 Database Systems and Data Analytics



# Objectives

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- Interacting with SQL Developer
- Connect to Oracle Server

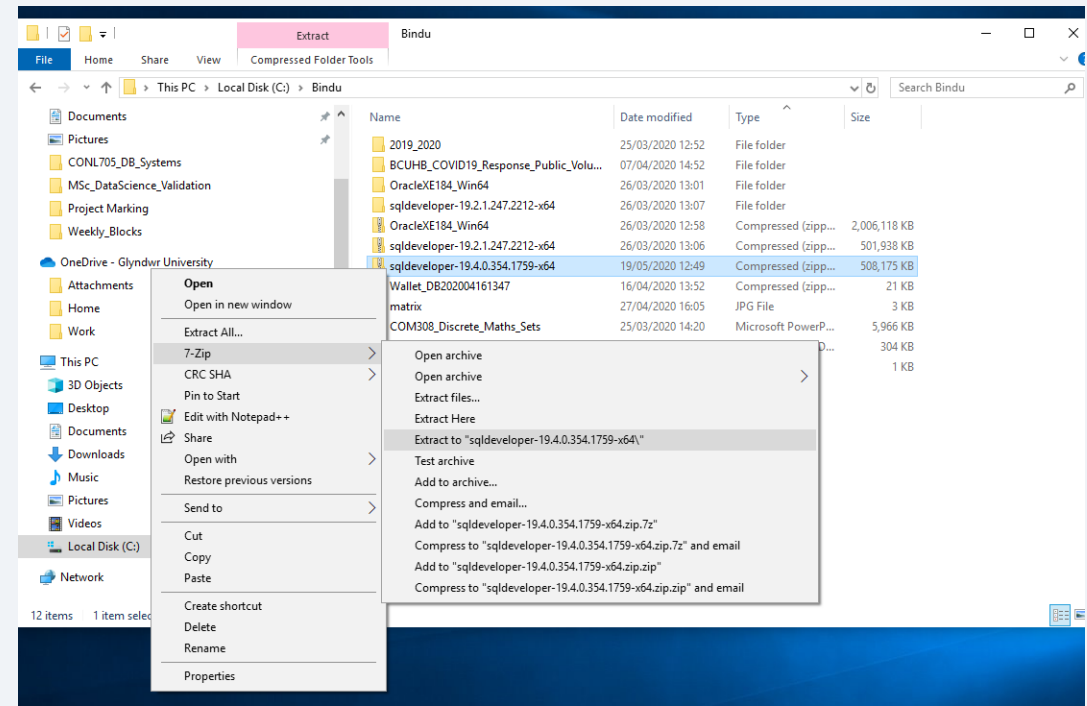
## Download SQL Developer and Cloud Wallet

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- **Download** the SQL Developer, the compressed file from Moodle
- Please ensure you have downloaded the version compatible to your operating system
  - Windows 64/32 with JDK
  - Mac OSX – Requires [JDK 8](#) or [11](#) required
  - Linux - [JDK 8](#) or [11](#) required
- **Download** 'Wallet\_\_\_\_\_ .zip' to your hard disk

# Unzip SQL Developer

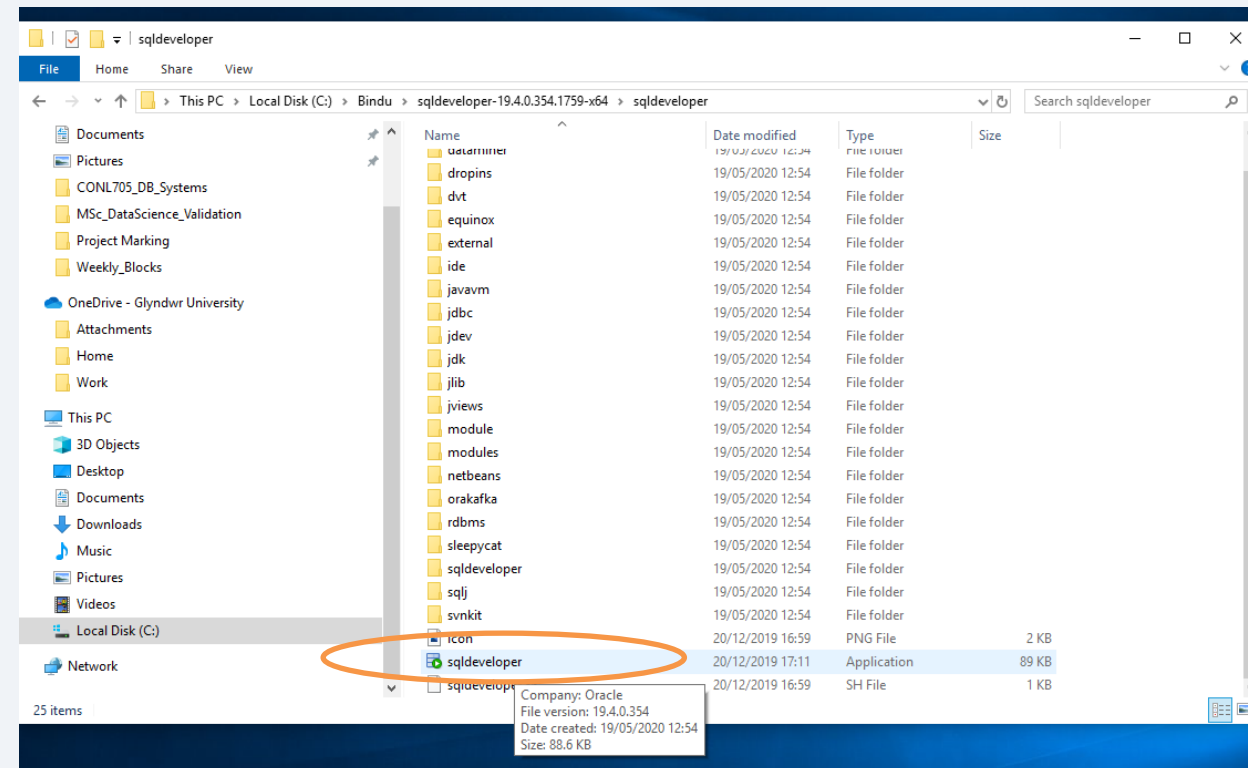
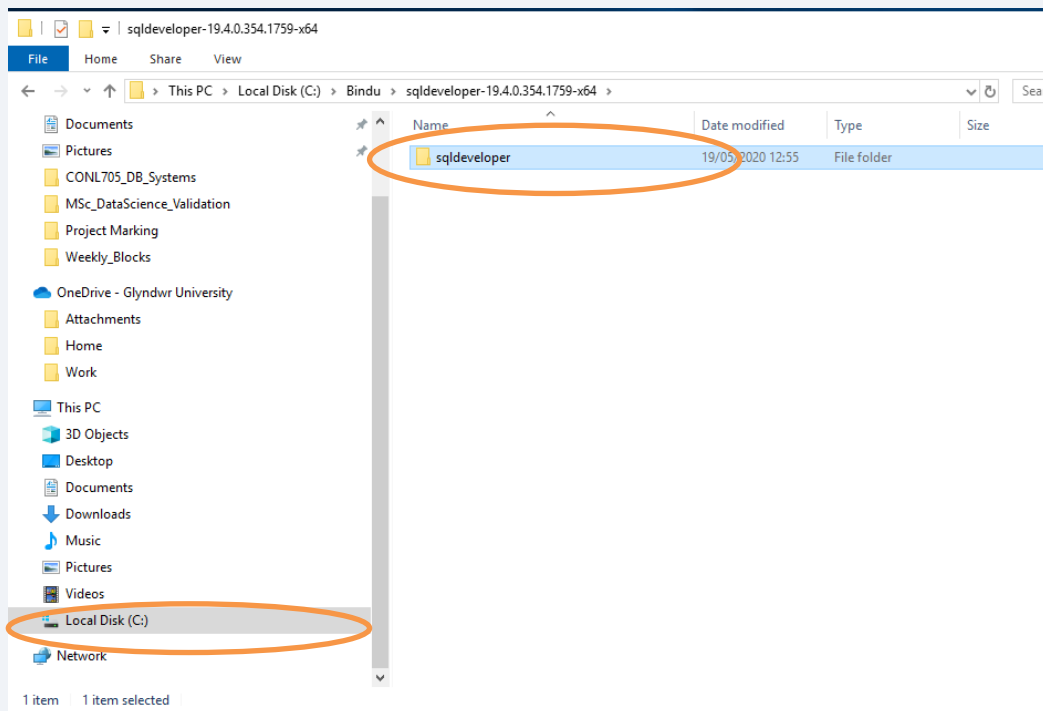
- Using an appropriate software **Unzip** the zip file to a folder into your hard disk
- For example in Windows 10 you can use Winzip or 7zip (<https://www.7zip.org/download.html>)



# Run SQL Developer Application

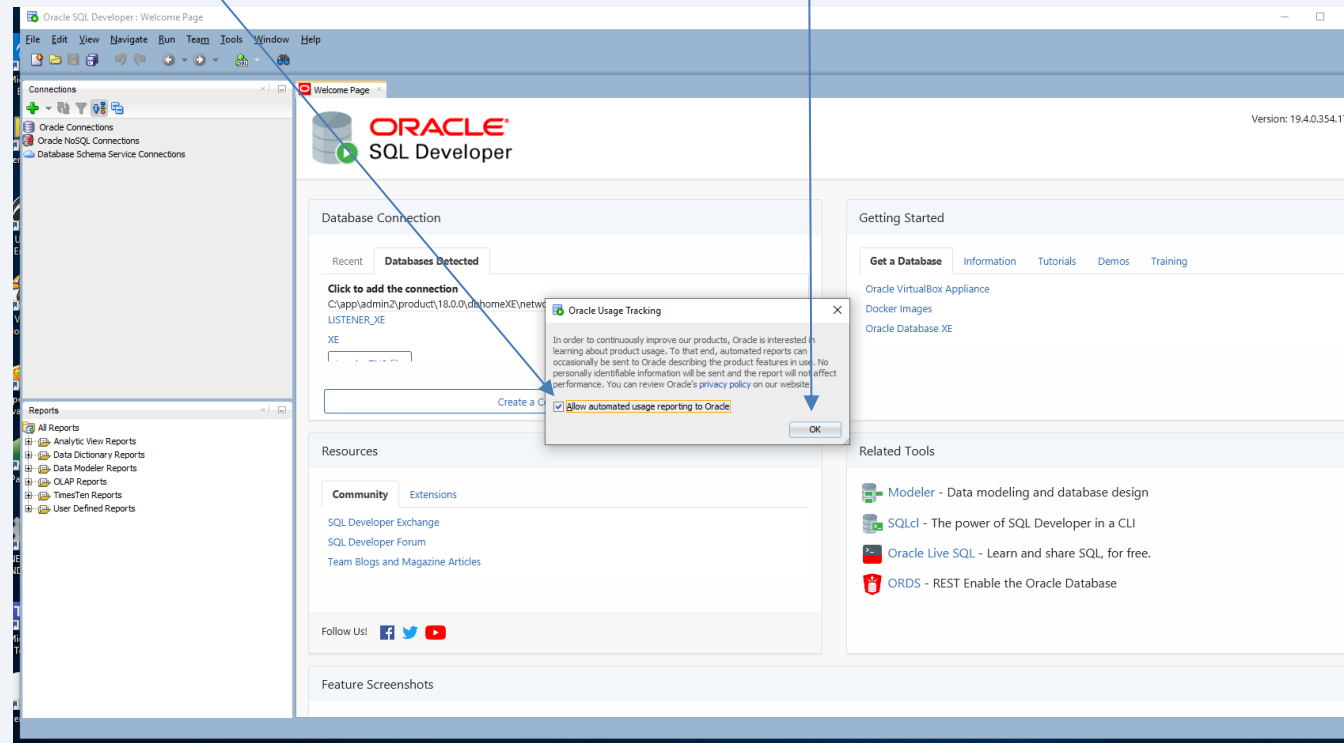
- Browse to the location of the SQL developer Application
- Double click o...

|              |                  |             |       |
|--------------|------------------|-------------|-------|
| sqldeveloper | 04/09/2019 23:25 | Application | 89 KB |
|--------------|------------------|-------------|-------|



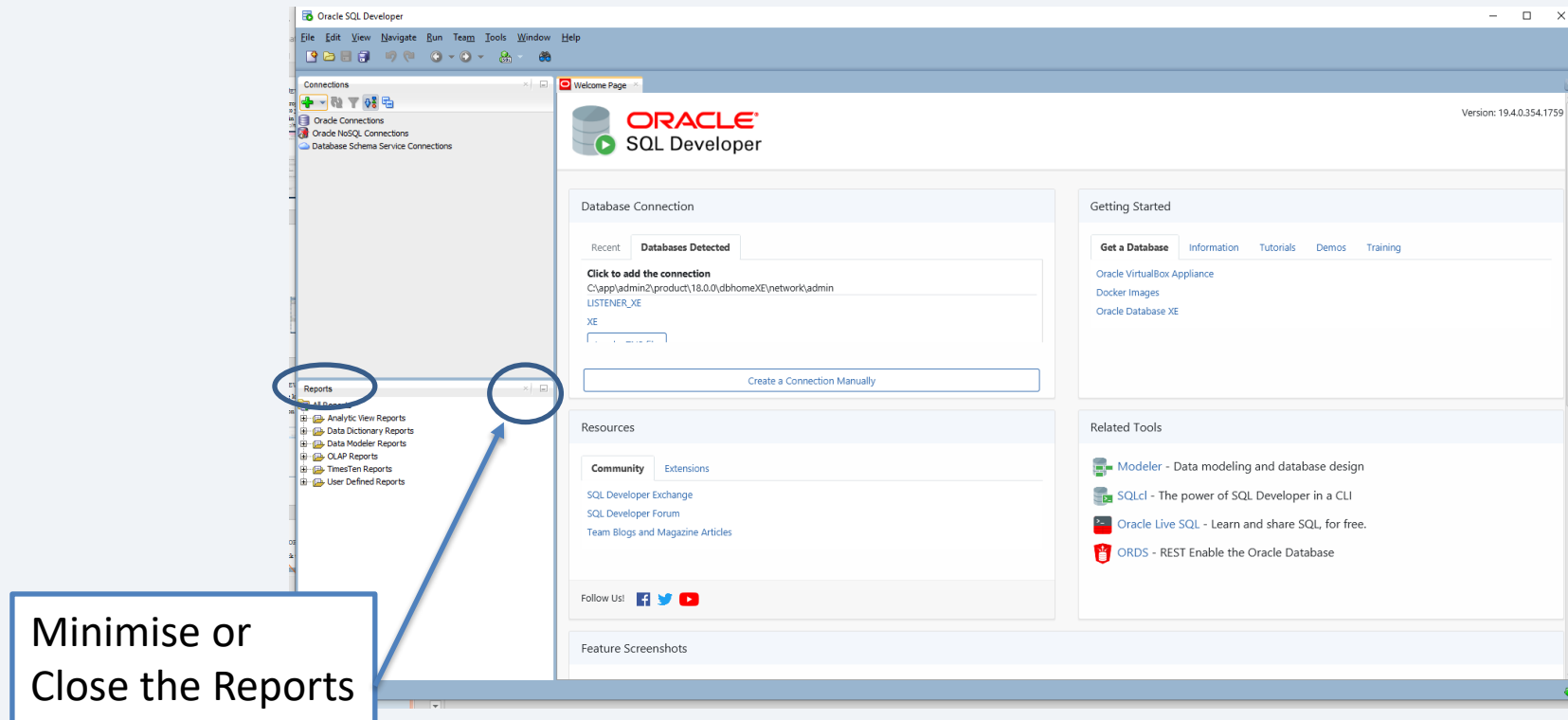
# The SQL Developer Interface

- Tick or Untick the box as you like, Click **OK**



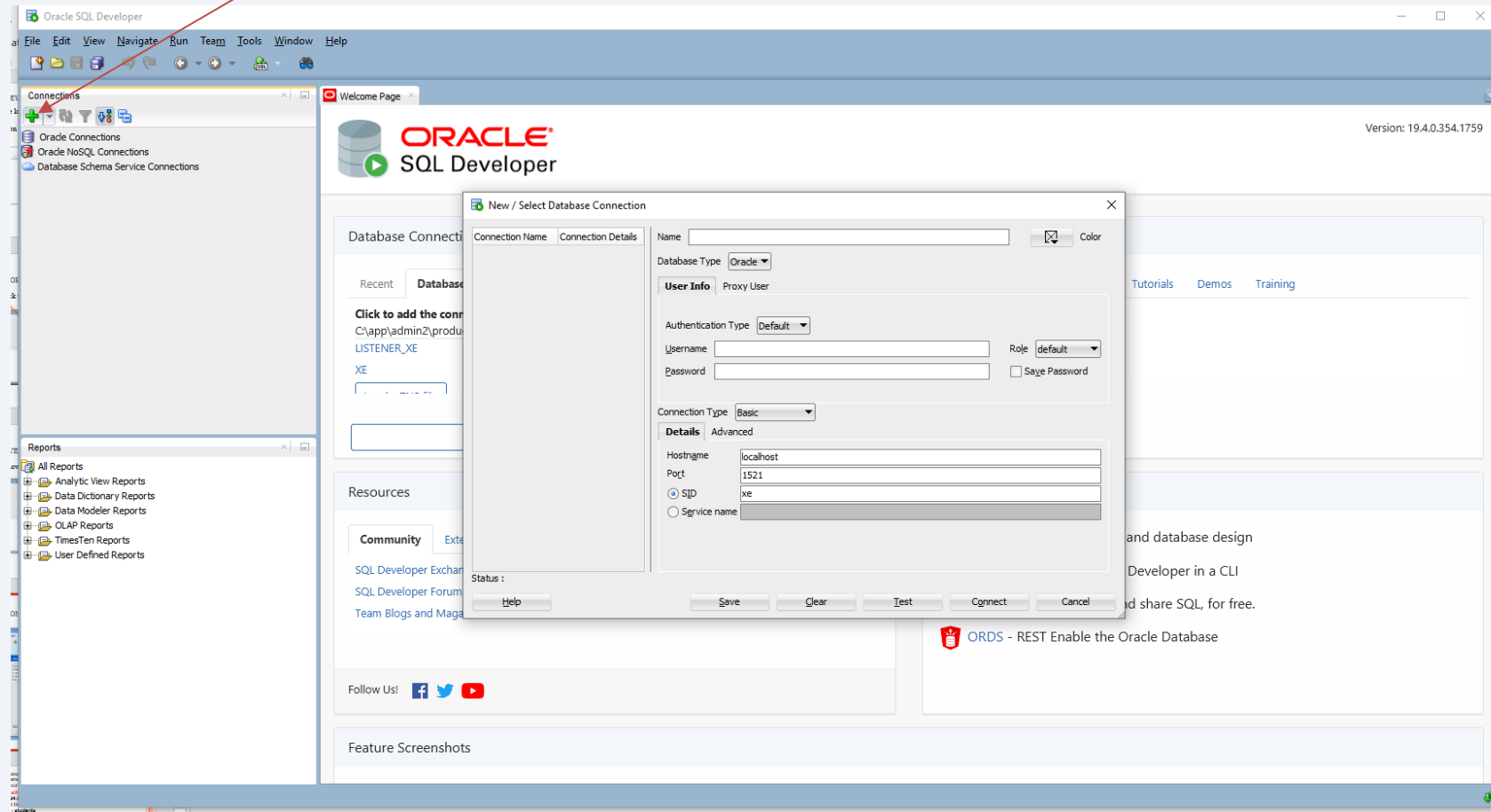
# SQL Developer – Welcome Screen

- You should now see the Welcome Screen



# Connecting to the Server: Add New Connection

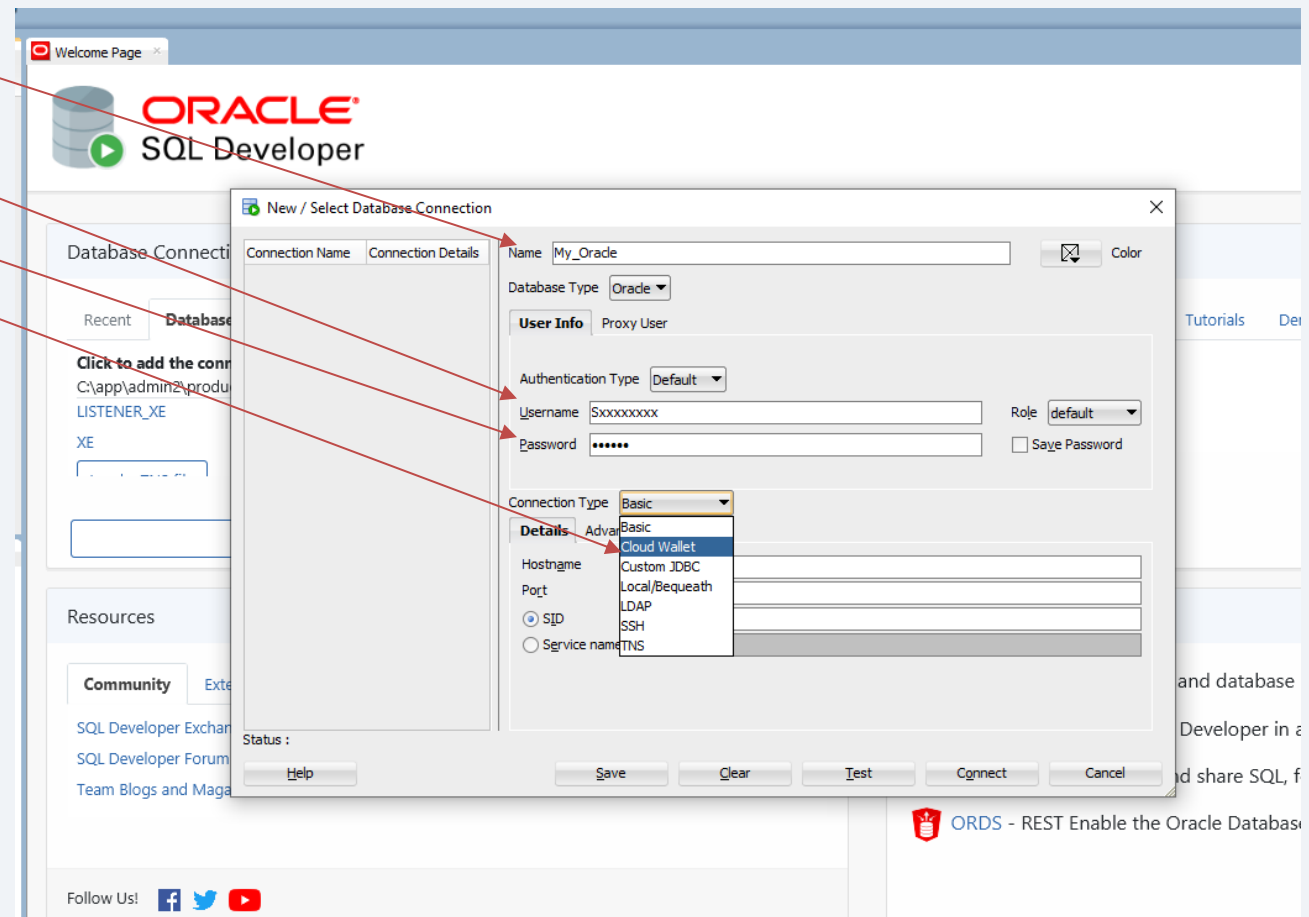
- Click on Add New Connection (+ Symbol)





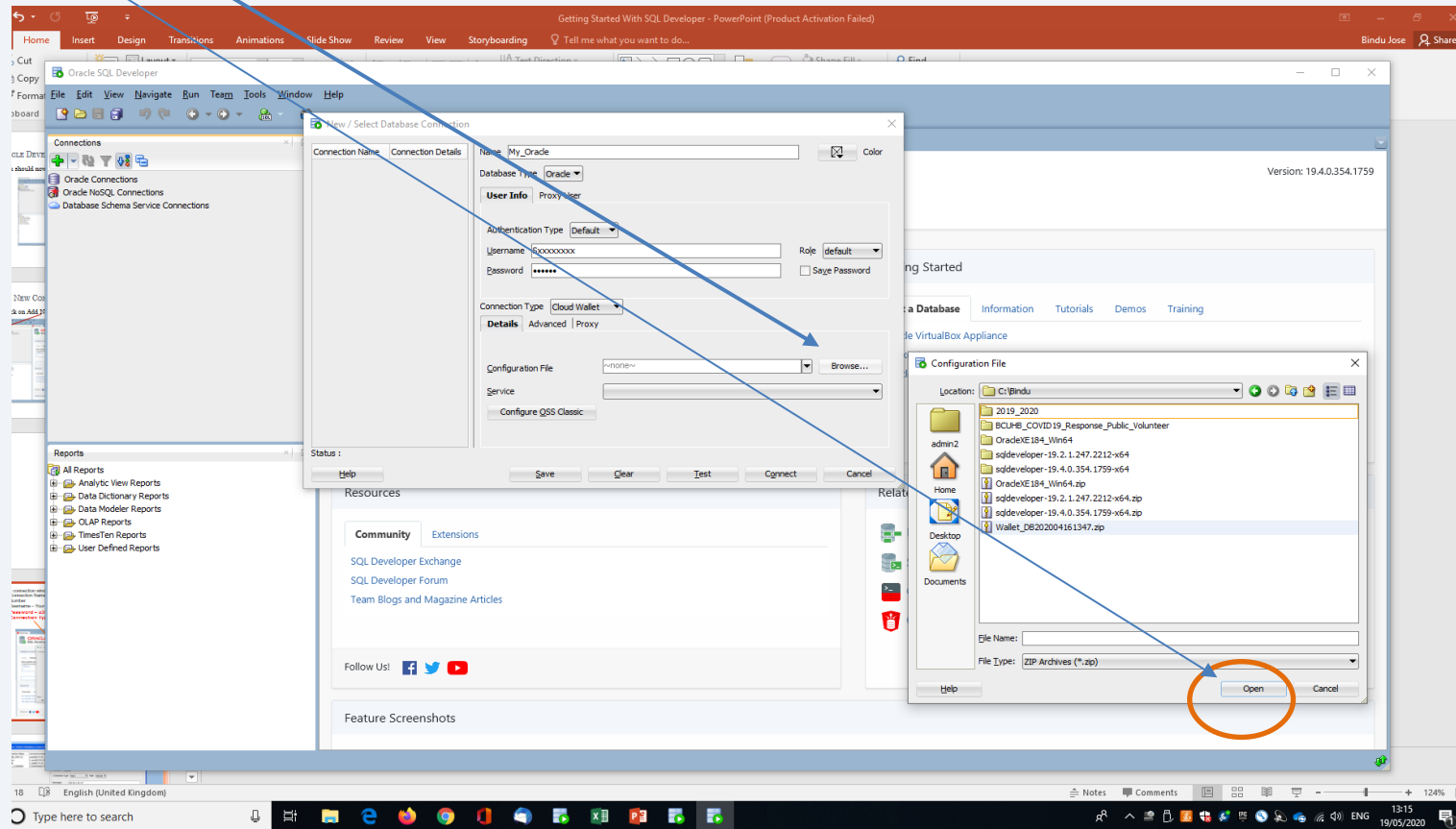
## Connection Details To Oracle Cloud Database Server

- In the connection window Type the following
  - Connection Name – can be any thing, preferably your Name or Student Number
  - Username – Your Student Number Sxxxxxxx
  - Password – COM438u30507
  - Connection Type – Cloud Wallet



# Cloud Wallet – Connection credentials

- Click **Browse** and then to the location you have saved 'Wallet\_DB2020\_\_\_\_.zip'
- Click **'Open'**

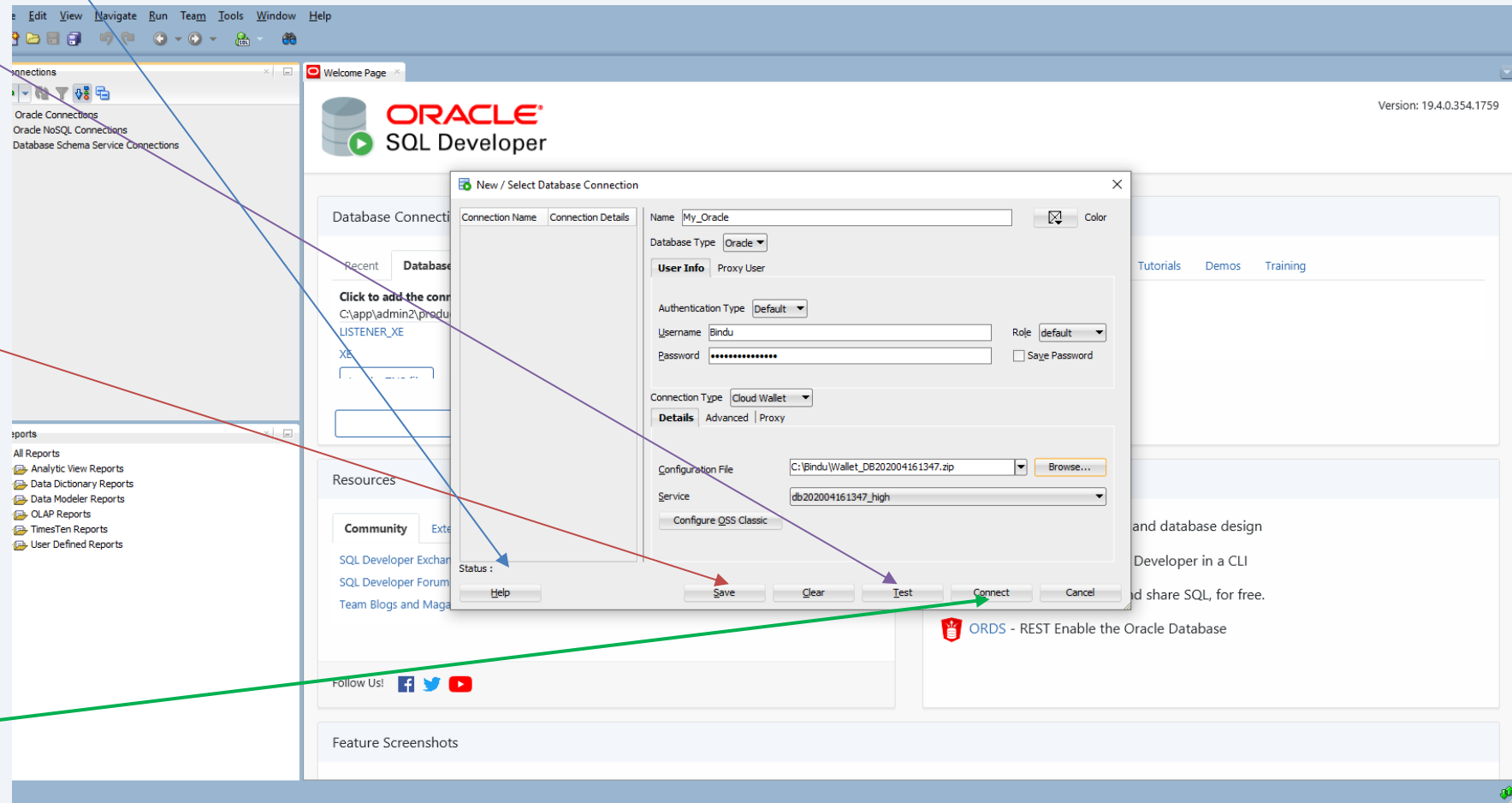


# Test, Save and Connect

- Once you filled the details –
  - Test** - If show **Success** (in case of failure check the connection credentials)

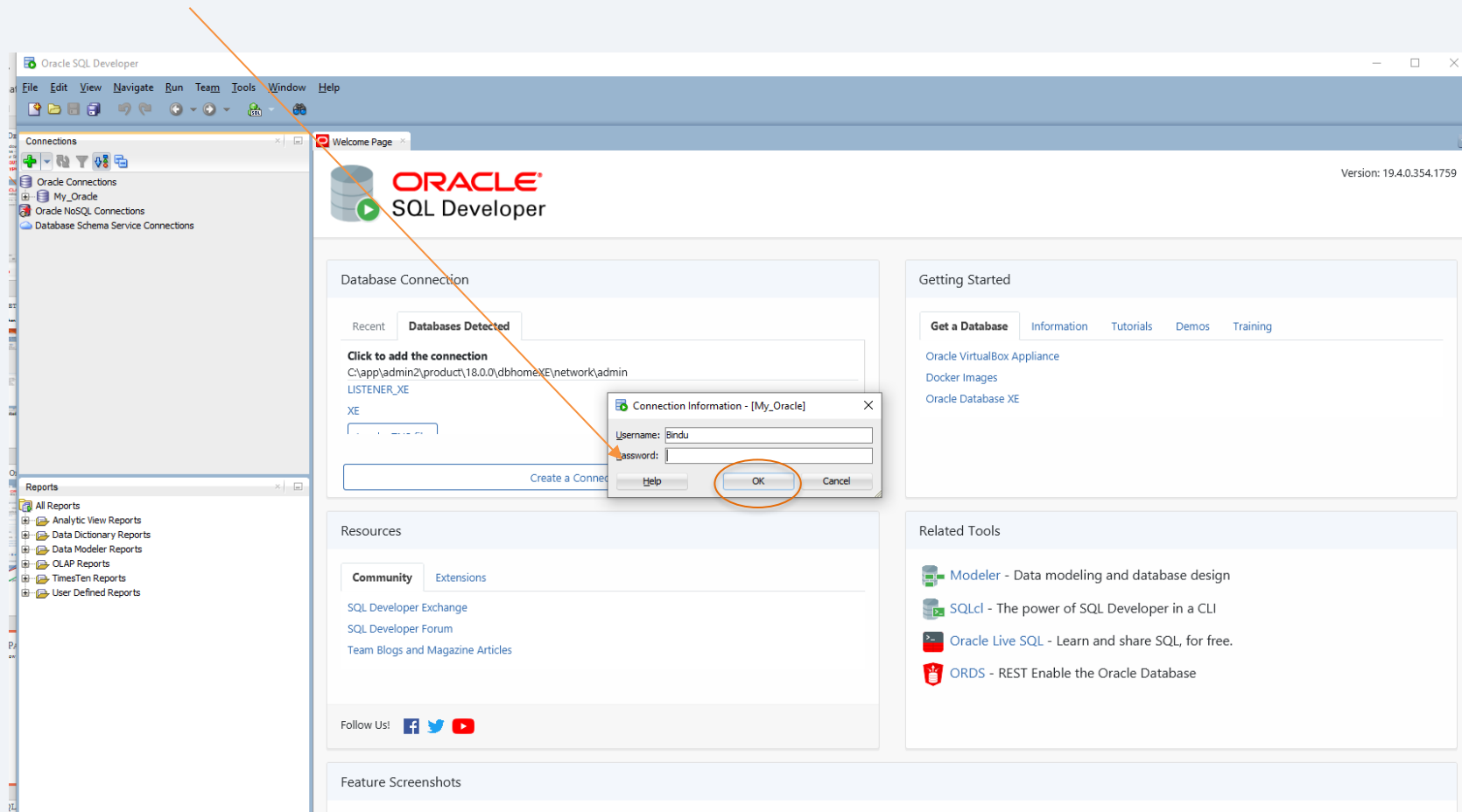
**Save**

**Connect**



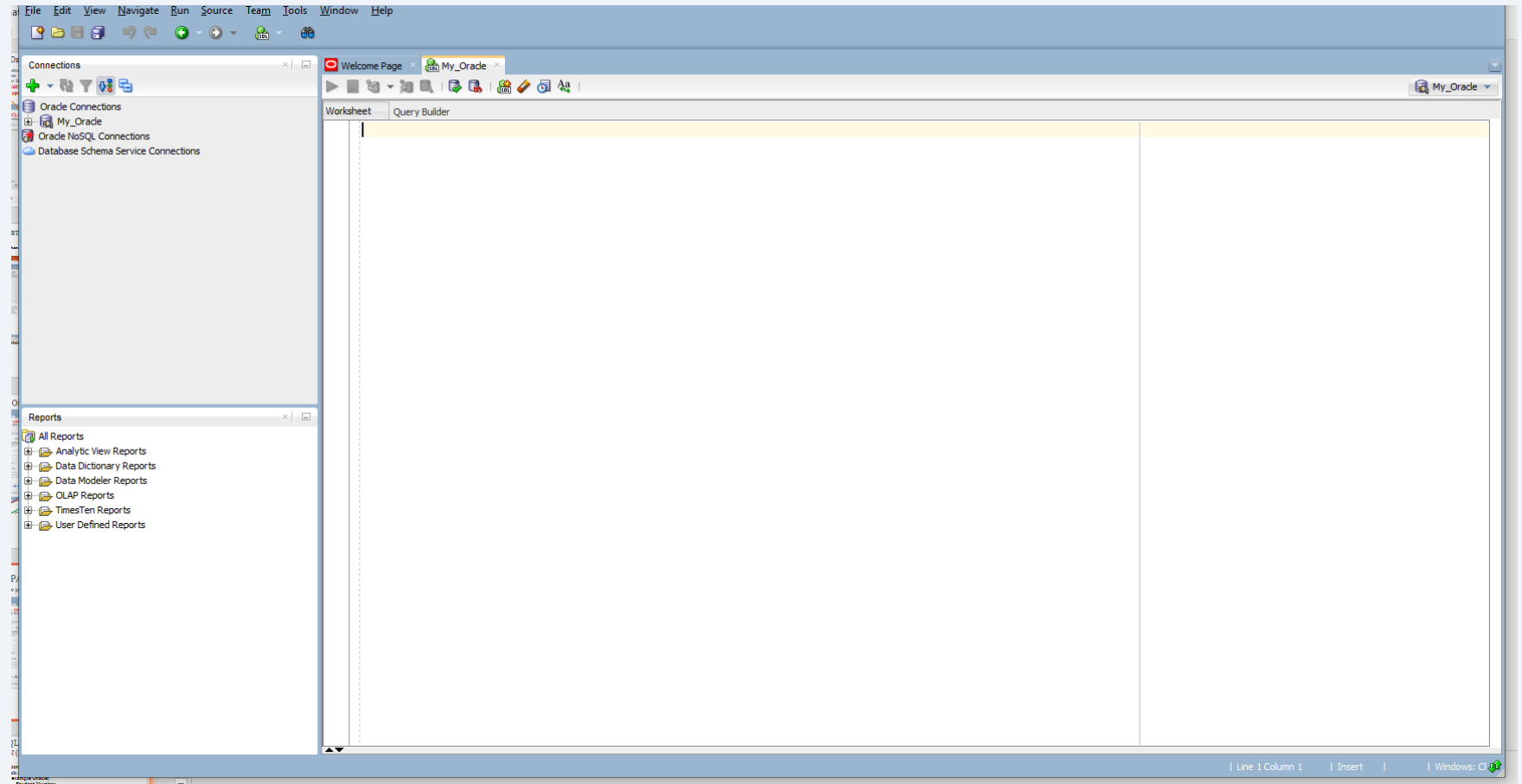
# Connect with the Password

- Connect will show you this screen:
- Enter the **password** and click **OK**



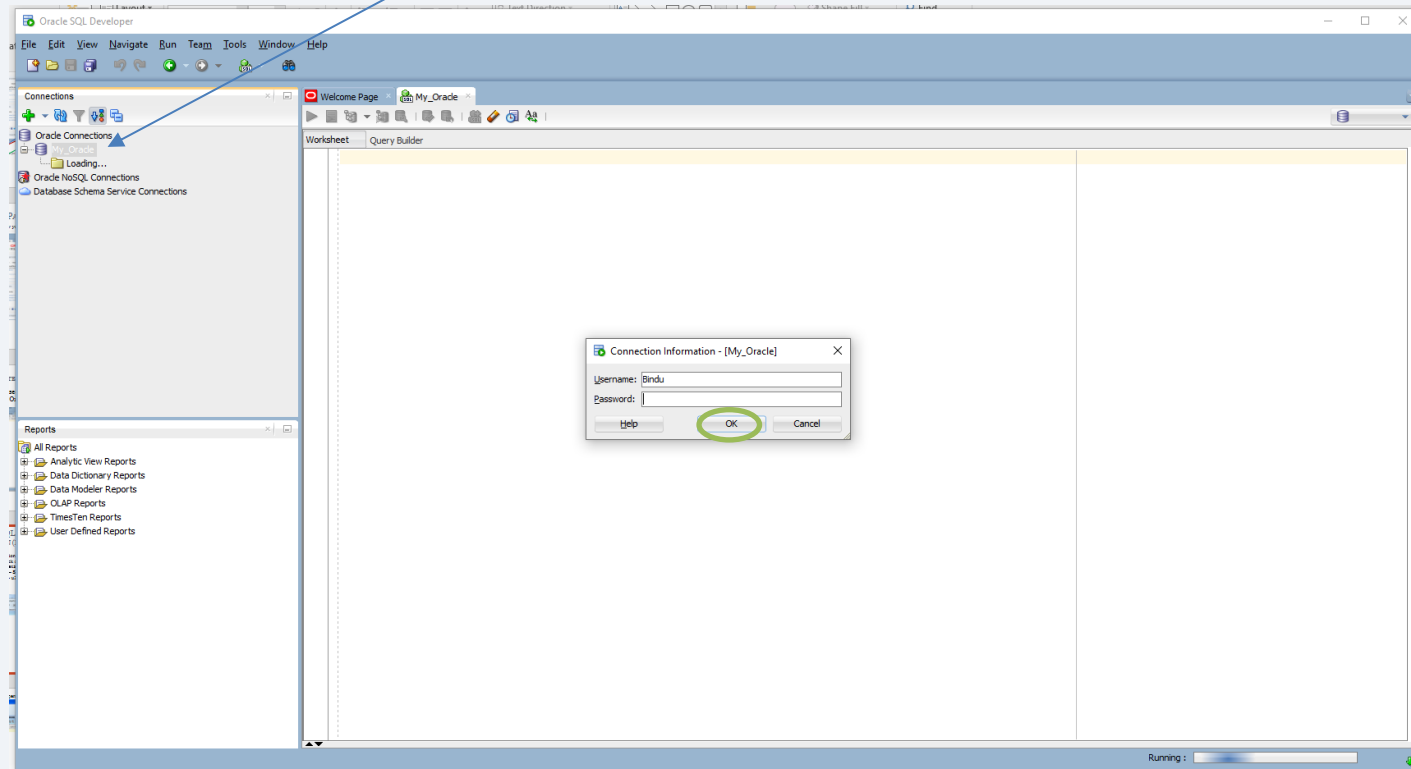
# Ready to Interact with Oracle Server

- You will now see the SQL worksheet and ready to interact with Oracle server.



# Login to SQL Developer Using an Existing Connection (if any)

- Under Connections
  - Double Click on Connection Name (for example My\_Oracle)
  - Username – **Student Number**
  - Password – **COM736u30507**
  - Click - **OK**



# Executing SQL Statements

The screenshot shows the Oracle SQL Developer interface with the following annotations:

- Execute **Single** Statement Ctrl+Enter**: Points to the Run button (a green play icon) in the toolbar.
- Execute **Multiple** Statements (Run Script F5)**: Points to the Run Script button (a green play icon with a document) in the toolbar.
- Clear Statements Ctrl+D**: Points to the Clear button (an eraser icon) in the toolbar.
- Yellow bar indicating active line or statement**: Points to the yellow highlight on the first line of the SQL statement in the worksheet.

The worksheet contains the following SQL statement:

```
1 SELECT * FROM EMP;
```

# Results Screen

The screenshot displays the Oracle SQL Developer interface. The title bar reads 'Oracle SQL Developer : JOSEB\_ORA11G'. The menu bar includes File, Edit, View, Navigate, Run, Versioning, Tools, and Help. The toolbar contains various icons for file operations and database actions. On the left, the 'Connections' pane shows a tree view with 'JOSEB\_ORA11G' selected. The main workspace is divided into two panes: 'Worksheet' and 'Query Builder'. The 'Worksheet' pane contains a single line of SQL code: `SELECT * FROM EMP;`. The 'Query Result' pane at the bottom displays the results of the query, showing 12 rows fetched in 0.094 seconds. The results are presented in a table with columns: EMPNO, ENAME, JOB, MGR, HIREDATE, SAL, COMM, and DEPTNO. The data includes employees like ALLEN, WARD, JONES, MARTIN, BLAKE, CLARK, KING, and TURNER. A red circle highlights the 'Run' button (a green play icon) in the toolbar. Two blue arrows point from text labels to the interface: one to the 'Script Window' (the Worksheet pane) and another to the 'Result Window' (the Query Result pane).

Script Window

Result Window

|    | EMPNO | ENAME  | JOB       | MGR    | HIREDATE  | SAL    | COMM   | DEPTNO |
|----|-------|--------|-----------|--------|-----------|--------|--------|--------|
| 1  | 7369  | (null) | CLERK     | 7902   | 17-DEC-80 | 800    | (null) | 20     |
| 2  | 7499  | ALLEN  | SALESMAN  | 7698   | 20-FEB-81 | 1600   | 300    | 30     |
| 3  | 7521  | WARD   | SALESMAN  | 7698   | 22-FEB-81 | 1250   | 500    | 30     |
| 4  | 7566  | JONES  | MANAGER   | 7839   | 02-APR-81 | 3272.5 | (null) | 20     |
| 5  | 7654  | MARTIN | SALESMAN  | 7698   | 28-SEP-81 | 1250   | 1400   | 30     |
| 6  | 7698  | BLAKE  | MANAGER   | 7839   | 01-MAY-81 | 3135   | (null) | 30     |
| 7  | 7782  | CLARK  | MANAGER   | 7839   | 09-JUN-81 | 2695   | (null) | 10     |
| 8  | 7839  | KING   | PRESIDENT | (null) | 17-NOV-81 | 5000   | (null) | 10     |
| 9  | 7844  | TURNER | SALESMAN  | 7698   | 08-SEP-81 | 1500   | 0      | 30     |
| 10 | 7876  | (null) | CLERK     | 7788   | 12-JAN-83 | 1100   | (null) | 20     |
| 11 | 7900  | (null) | CLERK     | 7698   | 03-DEC-81 | 950    | (null) | 30     |
| 12 | 7934  | (null) | CLERK     | 7782   | 23-JAN-82 | 1300   | (null) | 10     |



## Error Message: If you have no EMP table

The screenshot displays the SQL Developer environment. On the left is the 'Connections' tree with 'My\_Orade' selected. The main workspace is divided into three panes: 'Worksheet' (top), 'Script Window' (middle), and 'Result Window' (bottom). The 'Worksheet' pane contains the SQL query `SELECT * FROM EMP;`. A red arrow points to the 'Script Window' pane, which contains the error message: `ORA-00942: table or view does not exist`. A blue arrow points to the 'Result Window' pane, which is empty. A red box highlights the error message in the 'Script Window' pane, and a blue box highlights the 'Result Window' pane.

**Right Click and Select Toggle Line Numbers To see Line Number**

**Script Window**

**Error message**

**Result Window**

File Edit View Navigate Run Source Team Tools Window Help

Connections

Oracle Connections

My\_Orade

Tables (Filtered)

Views

Indexes

Packages

Procedures

Functions

Operators

Queues

Queues Tables

Triggers

Types

Sequences

REST Data Services

Materialized Views

Materialized View Logs

Synonyms

Public Synonyms

Database Links

Public Database Links

Directories

Editions

Application Express

Java

XML Schemas

XML DB Repository

OLAP Option

Analytic Views

Scheduler

RDF Semantic Graph

Recycle Bin

Other Users

Oracle NoSQL Connections

Database Schema Service Connections

Welcome Page My\_Orade My\_Orade~1

Worksheet Query Builder

SELECT \* FROM EMP;

Script Output Query Result

Executing: SELECT \* FROM EMP in 0 seconds

ORA-00942: table or view does not exist

00942. 00000 - "table or view does not exist"

\*Cause:

\*Action:

Error at Line: 1 Column: 16

Line 1 Column 20 Insert Modified Windows: C

# Test the SQL: Create Table

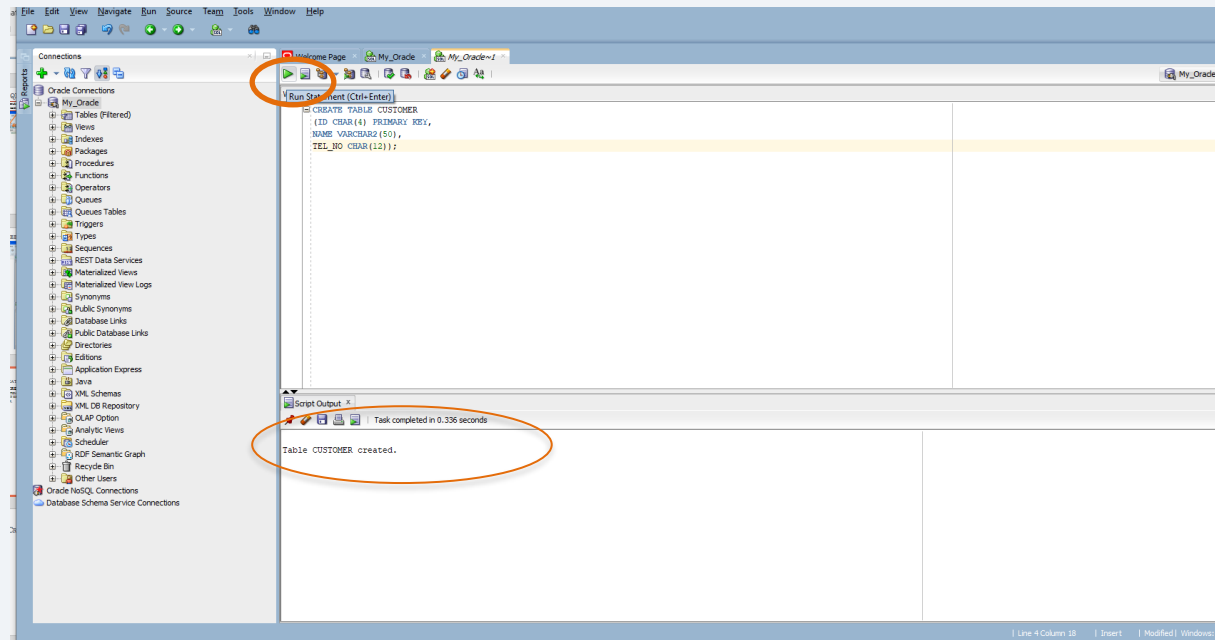
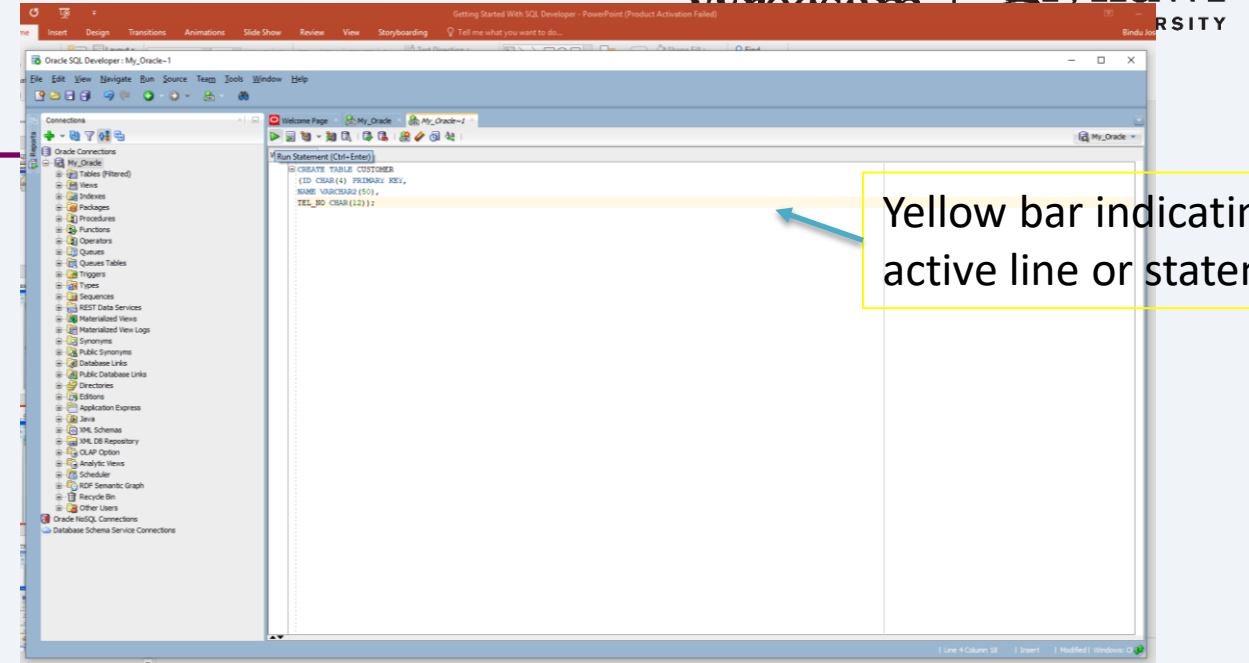
- Test the SQL for **creating table**

CREATE TABLE CUSTOMER

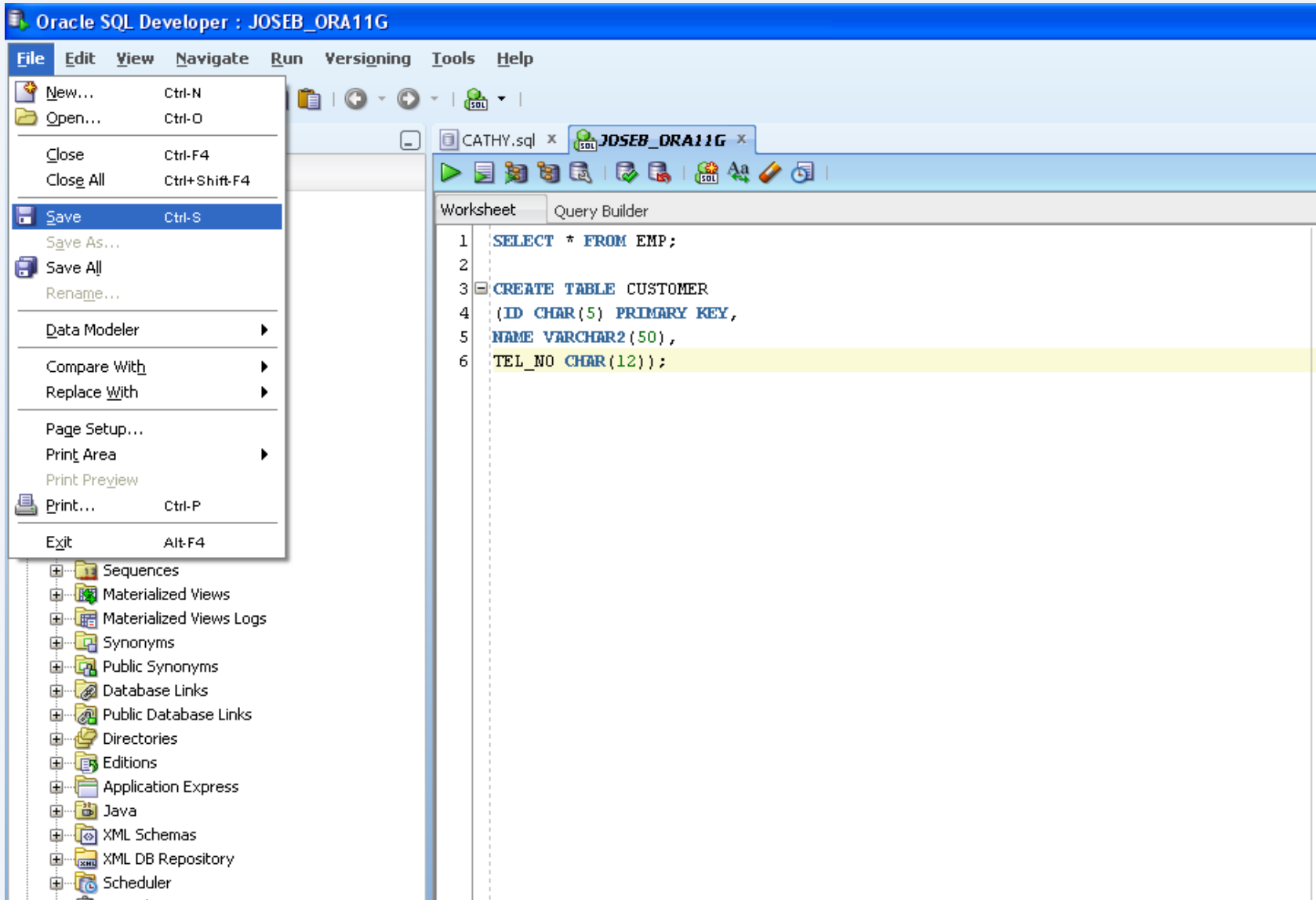
(ID CHAR(4) PRIMARY KEY,

NAME VARCHAR2(50),

TEL\_NO CHAR(12));



# Saving SQL Statements to a Script File



# Browse to the Location, Give a Name

Oracle SQL Developer : JOSEB\_ORA11G

File Edit View Navigate Run Versioning Tools Help

Connections x CATHY.sql x JOSEB\_ORA11G x

Worksheet Query Builder

```
1 SELECT * FROM EMP;  
2  
3 CREATE TABLE CUSTOMER  
4 (ID CHAR(5) PRIMARY KEY,  
5 NAME VARCHAR2(50),  
6 TEL_NO CHAR(12));
```

Save

Location: C:\Documents and Settings\joseb\Application Data\SQL Developer

Desktop  
My Docume...  
joseb  
Christina\_D...  
Assignment...

3 1/2 Floppy (A:)  
3 1/2 Floppy (B:)  
Local Disk (C:)  
Documents and Settings  
joseb  
Application Data  
SQL Developer  
DVD-RAM Drive (D:)  
Removable Disk (E:)  
Removable Disk (F:)  
Removable Disk (G:)  
Removable Disk (H:)  
USBDISKPRO (I:)  
Applications\$ on 'Fs2' (V:)  
StaffShare\$ on 'Fs1' (X:)  
Group Drive\$ on 'Fs2' (Y:)  
joseb\$ on 'fs2' (Z:)

File name: GIVE\_NAME

File type: SQL Script (\*.sql)

Help Save Cancel

Browse to the Location

Give a Name

Click Save

# Summary

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- Download SQL Developer, the wallet and connect to Oracle Server in the Cloud
- Interacting with the SQL Developer
- Work Through the exercise sheet